Zhe Yang

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RESEARCH INTERESTS

Committed to developing versatile robots and multimodal agents for everyday tasks. Current research centers on long-horizon tasks as a stepping stone toward real-world deployment:

- **Policy Design & Learning:** Developing policies that enable agents and robots to plan and act using long-term memory, current observations, and interactive feedback.
- **Benchmarking & Evaluation:** Constructing realistic environments and benchmarks to assess these capabilities in current agents and identify performance bottlenecks.

EDUCATION

Nanyang Technological University (NTU) Sep 2025 – Present
M.Sc. in Artificial Intelligence (MSAI) GPA: **5.0/5.0**
MMLab@NTU, advised by Prof. Chen Change Loy and Prof. Ziwei Liu

The Chinese University of Hong Kong (CUHK) Sep 2021 – May 2025
B.Sc. in Computer Science, *First Class Honours* Major GPA: **3.80/4.0**
ARISE Lab, advised by Prof. Michael R. Lyu

PUBLICATIONS

Peer Reviewed * Equal contribution

[1] HippoCamp: Benchmarking Contextual Agents on Personal Computers
Zhe Yang, S. Tian, K. Hu, S. Liu, H.-N. Nguyen, Y. Zhang, Z. Guo, M. Yu, Z. Zhang, J. Yang, C. C. Loy, and Z. Liu
*European Conference on Computer Vision (**ECCV**)*, 2026, pp. 98–118.

[2] Understanding Secret Leakage Risks in Code LLMs: A Tokenization Perspective
M. Chen, **Zhe Yang**, N. Huang, Y. Huang, Y. Li, Z. Li, and M. R. Lyu
*Findings of the Association for Computational Linguistics: **ACL 2026***, pp. 108–119.

[3] Data Compressibility Quantifies LLM Memorization
Y. Huang, **Zhe Yang**, M. Chen, N. Huang, J. Zhang, and M. R. Lyu
*Transactions on Machine Learning Research (**TMLR**)*, 2026.

[4] RAUIE: A Relation-Augmented Document-Level Event Extraction Model Based on UIE
Y. Liang, H. Jia, L. Wang, Y. Huang, and **Zhe Yang**
*5th International Seminar on Artificial Intelligence, Networking and Information Technology (**AINIT**)*, 2024.

[5] EgoTools: Towards Tool-Centric Reasoning in Real-World Egocentric Videos
S. Tian*, J. Kim*, S. Liu*, H. Li*, Y. Shen, S. Li, **Zhe Yang**, Y. Kim, F. Li, J. Wu, Y. Zhang, W. Wang, R. Yao, Y. Dong, Z. Chen, F. Hong, A. Furnari, J. Yang, H. Zhu, and Z. Liu
*Conference on Empirical Methods in Natural Language Processing (**EMNLP**)*, 2026. Accepted.

Manuscripts under Review

[6] InnerFeel: Benchmarking Multi-Perspective Emotion Report Prediction in Egocentric Social Interactions
X. Zhang*, S. Tian*, **Zhe Yang**, K. Zhu, Z. Huang, K. Hu, Y. Zheng, E. Cambria, and Z. Liu
Manuscript under review, 2026.

[7] FileGram: Grounding Agent Personalization in File-System Behavioral Traces
S. Liu, S. Tian, K. Hu, Y. Dong, **Zhe Yang**, B. Li, J. Yang, C. C. Loy, and Z. Liu
Manuscript under review, 2026. arXiv:2604.04901.

Technical Reports

[T1] Has GPT-6-Astra Achieved Embodied Intelligence?
Joint Research Team
Technical Report, 2026. arXiv:2610.22471.
Zhe Yang: Co-first author (alphabetical author order); benchmark prototype.

RESEARCH PROJECTS

Endogenous & Exogenous Memory for World Action Models

Simulation, Benchmark & Real-Robot Evaluation | Mentor: Dr. Zhonghua Wu Jun 2026 – Present

- **Dual-Memory Architecture:** Assisted in designing a dual-memory mechanism: endogenous memory compresses older history into gist tokens, while exogenous memory records regions where the model's predictions disagree with new observations, to handle external changes.
- **Benchmark & Simulation Validation:** Co-developed the RMBench++ memory benchmark, running training and simulation experiments to test how short- and long-term history jointly support decisions.
- **Real-Robot Pipeline Deployment:** Engineered the end-to-end real-robot pipeline from scratch, connecting data acquisition, model deployment, and robot control and debugging.

- **Interim Evaluation Results:** With endogenous memory alone, the model matches or exceeds MemoryWAM on all five RMBench tasks evaluated so far.

Native Latent Action Models: Robot Policy Learning from Action-Free Videos

Downstream Validation & Real-Robot Experiments | Mentor: Dr. Qingyi Tao Jun 2026 – Present

- **Model Integration & Simulation Evaluation:** Integrated register-token latent actions into the $\pi_{0.5}$ policy and ran closed-loop evaluation on manipulation tasks in the RoboTwin simulator.
- **Real-Robot Experiments:** Built the end-to-end real-robot experiment pipeline, covering teleoperated data collection, data formatting, model inference and deployment, and success-rate evaluation.
- **Key Advantage:** Extracts latent actions natively inside a pretrained video diffusion transformer, compressing its dense video features $\sim$1,700$\times$ (to 32$\times$128) so that capacity concentrates on action-relevant dynamics rather than video appearance; ranks first on CALVIN action regression among compared representations.

Probing GPT-6-Astra & Prototyping a Next-Generation Embodied Benchmark

Core Contributor: Benchmark Prototype & BEHAVIOR-1K Evaluation Sep – Oct 2026

- **Action Interface Redesign:** Identified the action interface as the key bottleneck on BEHAVIOR-1K and drove the switch from joint-space to end-effector-pose control, raising success from 1% to 37%, 2.5$\times$ the prior state of the art.
- **Benchmark Prototype Development:** Designed a next-generation embodied AI benchmark prototype on three principles: tasks rooted in human needs, no in-domain demonstrations, and a world clock that keeps running during agent deliberation.
- **Physics-Grounded Simulation:** Built a physics-grounded apartment in NVIDIA Isaac Sim with need-based household tasks judged on both explicit requests and unstated obligations (e.g., turning off the stove).
- **Failure Identification:** Co-ran the evaluation, revealing failures hidden by standard benchmarks: GPT-6-Astra lost track of its own changes (e.g., leaving the fridge open for 7+ min) and acted on observations that went stale while it deliberated.

Benchmarking Contextual Agents on Personal Computers (HippoCamp)

Research Lead & First Author | Advisors: Prof. Ziwei Liu & Prof. Chen Change Loy Sep 2025 – Mar 2026

- **Novel Evaluation Setting:** Defined an underexplored setting to assess whether agents can search, perceive, and reason over user device-scale multimodal personal files.
- **Device-Scale Environment Construction:** Built three archetypal long-lived environments from 100+ user interviews with pseudonymization and privacy filtering, spanning 2,000+ real files (42.4 GB) in heterogeneous formats across five modalities.
- **Step-wise Failure Diagnosis:** Curated 581 QA pairs and 46.1K structured trajectory annotations, enabling failures to be localized to search, perception, or reasoning stages.
- **Gap Analysis:** Developed the evaluation pipeline and benchmarked 9 methods spanning RAG, search agents, and autonomous agents; even the strongest, ChatGPT Agent Mode, reached only 48.3% on user profiling vs. 87% for humans, bottlenecked by multimodal perception and evidence grounding.
- **Project Leadership & Coordination:** Led the project end-to-end as first author, coordinating the team through scenario definition, data construction, and evaluation.

Tool-Centric Reasoning in Egocentric Videos (EgoTools)

Data Collection & Annotation | Advisor: Prof. Ziwei Liu Feb – Oct 2026

- **Data Platform & Collection:** Coordinated team collection of multimodal egocentric videos and independently built the data annotation platform to support large-scale annotation.
- **Data Curation & Annotation:** Carried out large-scale curation and annotation and co-designed the annotation scheme, helping build a 100-hour corpus (646 videos, 7 domains, 361K captions, 6.5K narrations).
- **Collective Evaluation Results:** Used the processed data, together with the team, to build a 1,000-question diagnostic benchmark, where Gemini-3.1-Pro reached 66.9% vs. 83.2% for humans, and fine-tuning Qwen3-VL-8B on our data raised accuracy from 50.0% to 60.9%.

Multi-Perspective Emotion Understanding in Social Interactions (InnerFeel)

Research Framing & Evaluation Design | Advisor: Prof. Ziwei Liu Jul – Oct 2026

- **Research Framing:** Co-defined the topic and framing of social understanding as per-user belief tracking with memory isolation for multi-perspective emotion prediction.
- **Annotation & Evaluation Design:** Set the annotation dimensions and 4-/5-level rating scales for self- and partner-reports, and co-designed a controlled privileged-information test of whether models leak one person's private report into judgments about another.

- **Collective Findings:** Built a benchmark from 36 conversations (61 participants) with the team, revealing that models are largely blind to interpersonal misunderstandings, predicting agreement for up to 86% of pairs when only 44% actually agree.
- **Writing & Visualization:** Co-wrote the manuscript and independently produced all figures.

Agent Personalization from File-System Behavior (FileGram)

Data & Behavior Simulation | Advisors: Prof. Ziwei Liu & Prof. Chen Change Loy Jan – Mar 2026

- **Real-World Data Adaptation:** Adapted my first-author benchmark HippoCamp into 32 task workspaces (615 privacy-processed files) to ground simulated users in real data.
- **Simulation Data Engine:** Independently developed a computer-use simulation in which persona-driven agents execute tasks in sandboxed workspaces, yielding 640 trajectories with 20K atomic file operations.
- **Evaluation Outcomes:** Used these traces to help build the first file-system personalization benchmark with 4.6K questions and the FileGramOS memory architecture, which outperformed all 12 baselines (59.6% vs. 51.9%), while all methods dropped to single digits on real-world screen recordings.

Entropy–Memorization Law: Evaluating Memorization Difficulty of Data in LLMs

Core Discoverer and Pipeline Lead | Advisor: Prof. Michael R. Lyu | Undergraduate Thesis Project Jun 2024 – May 2025

- **Memorization Law Discovery:** Formulated and validated the Entropy–Memorization Linearity: entropy of single samples correlates only moderately with memorization ($r \approx 0.6$), but entropy over sets of samples tracks it near-linearly ($r = 0.92$–0.97), holding across four model families and most semantic domains.
- **Pipeline & Inference:** Engineered a 400K+ sample inference pipeline, finding that perfectly memorized text draws on only 1/50 of the vocabulary; co-developed a training-free dataset inference test, correct on 5/6 benchmark cases vs. 3/6 for a prior method, with at least 2× speedup.
- **Code Leakage Tracing:** Co-identified gibberish bias in code LLMs, where high-character-entropy secrets, such as API keys, become low-token-entropy sequences under BPE, increasing leakage risks.
- **Mitigation & Scaling:** Traced the bias to token-distribution shifts between code and secret data, showing that it grows with vocabulary size (39K–94K), and proposed character-level tokenization as mitigation.

INVITED TALKS & PRESENTATIONS

2nd Workshop on Benchmarking Evidence-Aligned Multimodal Reasoning (BEAM 2)

Invited Paper Presentation | ECCV 2026 8 Sep 2026

Title: *HippoCamp: Benchmarking Contextual Agents on Personal Computers*

INDUSTRY EXPERIENCE

SenseTime International Pte. Ltd., Singapore Jun 2026 – Present

Algorithm Intern, Embodied AI

- Core contributor to the GPT-6-Astra technical report (benchmark prototype; BEHAVIOR-1K evaluation).
- Research on endogenous and exogenous memory for world action models and on native latent action models for robot policy learning (see Research Projects).

GEOVIS Technology Co., Ltd. Jul – Aug 2024

Algorithm Intern

- Built an automated data collection framework for a multi-source data visualization platform, fine-tuning OneKE for unstructured data parsing to enable automated knowledge-graph generation.
- Designed and deployed a Neo4j-based question-answering backend over the generated knowledge graph, supporting knowledge queries and platform interaction.

AWARDS AND HONORS

HKSAR Government Scholarship (HK$80,000) 2024–2025

Dean's List (four times), Faculty of Engineering, CUHK 2021–2025

Silver Award for Outstanding Academic Performance, Dept. of CSE, CUHK May 2024